For M/M/1 queue:
Utilisation = arrival rate / processing rate

In integer programming or linear programming:
Utilisation =
Total amount of traffic on the link /
link capacity

E = {<1,2>, <2,1>, <1,3>, <2,4>, <4,2>, …}

Question 1: Does the following represent a valid path from source 1 to destination 6?

$x_{1,2} = 1$

$x_{2,5} = 1$

$x_{4,6} = 1$

with the rest of $x_{\,,\,}$ being zero.

Question 2: Does the following represent a valid path from source 1 to destination 6?

$x_{1,2} = 1$

$x_{2,5} = 1$

$x_{5,6} = 1$

with the rest of $x_{\,,\,}$ being zero.

Use constraints to ensure only valid paths are used.

At the source node, the constraint is:

$x_{1,2} + x_{1,3} = 1 + x_{2,1}$

For simplicity, let us assume $x_{2,1}$ is zero. We have:

$x_{12} + x_{13} = 1$

Since x_{12} and x_{13} can be either 0 or 1, there are only two possible combinations of x_{12} and x_{13} . What are they?

There are two possibilities:
Either
$x_{12} = 0$ and $x_{13} =1$; or

$x_{12} = 1$ and $x_{13} = 0$

Source node is Node 1

$$\sum_{j:\langle 1,j\rangle\in E} x_{1,j} - \sum_{j:\langle j,1\rangle\in E} x_{j,1} = 1$$

Destination node is Node 6

$$\sum_{j:\langle 6,j\rangle\in E} x_{6,j} - \sum_{j:\langle j,6\rangle\in E} x_{j,6} = -1$$

Nodes other than source and destination

$$\sum_{j:\langle i,j\rangle\in E} x_{i,j} - \sum_{j:\langle j,i\rangle\in E} x_{j,i} = 0, \;\; i \in N - \{s,d\}$$

Consider:

$6\ x <= 5$ and $x \in \{0,1\}$

What can x be?

Consider:

$5\ x <= 6$ and $x \in \{0,1\}$

What can x be?

- SPP can be formulated as

$$\min \sum_{\langle i,j\rangle \in E} c_{i,j} x_{i,j}$$

subject to

$$\begin{aligned}
\sum_{j:\langle i,j\rangle \in E} x_{i,j} - \sum_{j:\langle j,i\rangle \in E} x_{j,i} &= \begin{cases} 1 & \text{if } i = s \\ 0 & \text{if } i \in N - \{s, d\} \\ -1 & \text{if } i = d \end{cases} \\
x_{i,j} &\in \{0, 1\} \text{ for all } \langle i, j\rangle \in E
\end{aligned}$$

$$\min \sum_{(i,j)\in E} c_{ij} x_{ij}$$

subject to

$$\begin{aligned}
\sum_{j:(i,j)\in E} x_{ij} - \sum_{j:(j,i)\in E} x_{ji} &= \begin{cases} 1 & \text{if } i = s \\ 0 & \text{if } i \in N - \{s,d\} \\ -1 & \text{if } i = d \end{cases} \\
f x_{ij} &\leq b_{ij} \text{ for all } (i,j) \in E \\
x_{ij} &\in \{0,1\} \text{ for all } (i,j) \in E
\end{aligned}$$